

Phase 2

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1.The approach we used

The Approach we used is **Rule-Based Expert System (RBES)** Our system is designed as a **Rule-Based Expert System (RBES)**, which is a classic and reliable paradigm of Knowledge Based Systems. It operates by encoding clinical expertise into a formal **Knowledge Base** consisting of a library of **Production Rules**. These rules follow a logical **IF-THEN** structure, where the "IF" part represents a specific clinical pattern (symptoms + vitals), and the "THEN" part represents the inferred cardiac condition and its associated priority level.

2. So why this approach?

- **Mapping to Clinical Protocols:** Medical knowledge and heart failure management are inherently algorithmic and follow precise pathways. Since clinical triage follows standardized "Decision Trees," a Rule-Based approach is the most reliable method to translate these structured protocols into a digital format without compromising medical logic.
- **Medical Explainability and Transparency:** In critical healthcare, "Black-Box" models—where decisions are hidden—are insufficient. Our system provides an explicit **Audit Trail**. Because each decision is tied to a specific rule, the system can explain the "Why" behind its conclusion ("Priority is Critical because Heart Rate > 110 and Acute Chest Pain are present"). This transparency is vital for patient safety and professional trust.
- **Expert Knowledge via Clinical Thresholds:** This approach allows us to deliver specialist-level expertise directly to the user. By utilizing fixed **Clinical Measurement Thresholds** (such as Oxygen Saturation $< 90\%$ or Systolic BP ≥ 140), the system eliminates human subjectivity and ensures that every patient is evaluated against the same high-standard criteria used by experts 24/7.
- **Optimized for Telemedicine Environments:** Rule-Based systems are computationally efficient and do not require massive datasets or high-performance hardware. This makes it the perfect solution for a **Telemedicine context**, allowing the system to provide rapid, consistent, and evidence-based triage on mobile platforms, even in rural or underserved areas where quick decisions save lives.

3. Type of Reasoning: Forward Chaining

The system employs **Forward Chaining** (Data-Driven Reasoning). The process starts with the "Initial Facts" (the symptoms and vital signs entered by the patient). It then searches the Knowledge Base for rules where the **IF** part matches these facts. When a match is found, the rule is executed, and the **THEN** part (conclusion) is added to the system as a "New Fact." This cycle repeats until a final diagnosis and urgency level are reached.

4. Conceptual Decision-Making (How the system reaches decisions)

The system reaches decisions through a cyclical process:

1. **Fact Gathering (Working Memory):** Patient inputs (e.g., severe chest pain, high heart rate) are stored as initial facts.
2. **Rule Matching (Inference Engine):** The engine matches these facts against the Rule Base.
3. **Fact Evolution:** Triggered rules produce intermediate facts (ex "Potential Acute Event"). These new facts are fed back into the working memory.
4. **Final Conclusion:** The cycle terminates when a rule providing a final "Urgency Level" and "Suggested Action" is triggered.

Step-by-Step Reasoning Examples

Example 1: Moderate Case

- **Input Facts:** Shortness of breath = Yes, Edema = Yes, BP = 150/95.
- **Step 1:** Rule matches Shortness of breath + Edema --> Adds new fact: Fluid Congestion.
- **Step 2:** Rule matches Fluid Congestion + High BP --> Adds new fact: Diagnosis = Stable Chronic Heart Failure.
- **Step 3:** Rule matches Diagnosis --> Concludes Urgency = Moderate and Action = Visit doctor within days.

Example 2: Critical Case

- **Input Facts:** Chest Pain = Severe, Heart Rate = 115.
- **Step 1:** Rule matches Severe Chest Pain --> Adds new fact: Status = Potential Acute Event.
- **Step 2:** Rule matches Status + High Heart Rate --> Adds new fact: Diagnosis = Acute Myocardial Infarction.
- **Step 3:** Rule matches Diagnosis --> Concludes Urgency = Critical and Action = Go to the Emergency Room Immediately.

